Linear Systems of Differential Equations

Elimination, Eigenvalues, and Forced Systems

M3503 Ordinary Differential Equations • Unit 4
Department of Mathematics & Computer Science • National University of Lesotho

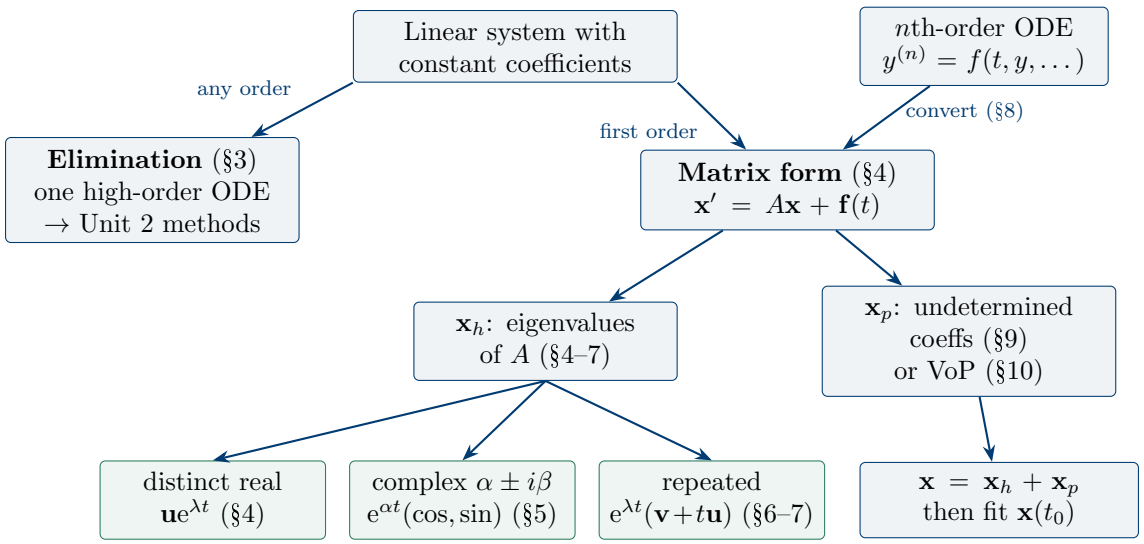
1. Why systems?

Most models do not come as one equation. Two tanks exchanging brine, a predator and its prey, the current in two coupled loops of a circuit, the position *and* velocity of a spring: each is a set of unknowns whose rates of change depend on one another. Such a set is a **system** of ODEs.

Units 1–3 gave us tools for a single equation. This unit extends them in two directions:

- **Elimination.** Remove unknowns one at a time until a single higher-order ODE remains, then use Unit 2.
- **Matrix methods.** Write the system as $\mathbf{x}' = A\mathbf{x}+\mathbf{f}(t)$ and let the eigenvalues and eigenvectors of A do the work.

> **The right analogy: one equation, many unknowns.** For $y' = ay$ the solution is Ce^{at}: the rate a sets the growth. For $\mathbf{x}' = A\mathbf{x}$ the "rates" are the eigenvalues of A, and the eigenvectors are the directions along which the system grows or decays like a single exponential. Diagonalise the coupling, and a coupled system falls apart into independent scalar pieces.



The left branch works for any constant-coefficient system, of any order. The right branch needs a first-order system, but scales to many unknowns and is the route used in applications and software.

2. Linear systems and operator notation

Linear system

A system of ODEs in unknowns $x_1(t), \ldots, x_n(t)$ is **linear** if it is linear in every x_j and in all their derivatives: no products x_1x_2, no powers x_1^2, no functions such as $\sin x_1$. It is **homogeneous** if every forcing term is zero, and **non-homogeneous** otherwise. In this unit all coefficients are constants.

The operator D. Write $D = \dfrac{d}{dt}$, so $x' = Dx$ and $x'' = D^2x$. A constant-coefficient expression such as $x'' - x$ becomes $(D^2 - 1)x$. Such operators multiply, factor and *commute* like polynomials: $(D-1)(D+1) = (D+1)(D-1) = D^2 - 1$. This is what lets us eliminate unknowns exactly as in simultaneous linear equations.

Example 1 — a second-order, non-homogeneous system

$$x'' + y' - x + y = -1, \qquad x' + y' - x = t^2.$$

This is a coupled second-order linear system with constant coefficients. In D-form:

$$(D^2 - 1)\,x + (D+1)\,y = -1 \quad (1), \qquad (D-1)\,x + D\,y = t^2 \quad (2).$$

3. The elimination method

Elimination reduces a system to a single higher-order ODE in one unknown — which Unit 2 already solves. It is the operator version of eliminating a variable from simultaneous equations.

Procedure

1. Write the system in D-form.
2. Apply suitable operators to the equations and add or subtract to eliminate one unknown.
3. Repeat until one equation in one unknown remains; solve it (complementary + particular).
4. Recover the other unknown(s), by eliminating again or by substituting back.
5. **Fix the constants.** Solving for each unknown separately creates too many arbitrary constants; substitute into an original equation to relate them.

How many constants? For $L_{11}x + L_{12}y = f_1$, $L_{21}x + L_{22}y = f_2$, the number of independent arbitrary constants equals the degree in D of the operator determinant

$$\Delta(D) = \begin{vmatrix} L_{11}(D) & L_{12}(D) \\ L_{21}(D) & L_{22}(D) \end{vmatrix}.$$

Compute it first, so you know how many constants the final answer must keep.

Example 2 — Example 1 solved

Solve $(D^2 - 1)x + (D+1)y = -1$ (1), $(D-1)x + Dy = t^2$ (2).
Constants. $\Delta(D) = D(D^2-1) - (D+1)(D-1) = (D^2-1)(D-1)$: degree 3, so **three** constants.
Eliminate y. Apply D to (1) and $-(D+1)$ to (2), and add. Since $D(-1) = 0$ and $(D+1)t^2 =$

$2t + t^2$,

$$\big[D(D^2-1) - (D+1)(D-1)\big]x = -t^2 - 2t \implies (D^2-1)(D-1)\,x = -t^2 - 2t. \qquad (3)$$

Complementary solution. $(m^2-1)(m-1) = 0$ gives $m = -1$ and $m = 1$ (double): $x_c = (c_1 + c_2 t)\mathrm{e}^t + c_3\mathrm{e}^{-t}$.
Particular solution. $m = 0$ is not a root, so try $x_p = A + Bt + Ct^2$. Since $(D^3 - D^2 - D + 1)x_p = (A - B - 2C) + (B - 2C)t + Ct^2$, matching $-2t - t^2$ gives $C = -1$, $B - 2C = -2 \Rightarrow B = -4$, $A - B - 2C = 0 \Rightarrow A = -6$.

$$x(t) = (c_1 + c_2 t)\mathrm{e}^t + c_3\mathrm{e}^{-t} - t^2 - 4t - 6.$$

Find y without new constants. Subtracting (2) from (1) in their original form gives $x'' - x' + y = -1 - t^2$, so $y = -1 - t^2 + x' - x''$. With $x' = (c_1 + c_2 + c_2 t)\mathrm{e}^t - c_3\mathrm{e}^{-t} - 2t - 4$ and $x'' = (c_1 + 2c_2 + c_2 t)\mathrm{e}^t + c_3\mathrm{e}^{-t} - 2$,

$$\boxed{x = (c_1 + c_2 t)\mathrm{e}^t + c_3\mathrm{e}^{-t} - t^2 - 4t - 6, \qquad y = -c_2\mathrm{e}^t - 2c_3\mathrm{e}^{-t} - t^2 - 2t - 3}$$

Three constants, as predicted. *Check* in (2): $x' - x + y' = t^2$ ✓.

Alternative route. From (2), $y' = t^2 + x - x'$; integrating gives y plus a new constant K. Substituting into (1) forces $K = -3$ — the same answer. Whenever you integrate, an original equation must be used to pin the extra constant.

Example 3 — when the constants must be related

Solve $x' - 4x + y'' = t^2$, $x' + x + y' = 0$. In D-form:

$$(D-4)\,x + D^2 y = t^2 \quad (3), \qquad (D+1)\,x + D\,y = 0 \quad (4).$$

Constants. $\Delta(D) = D(D-4) - D^2(D+1) = -(D^3 + 4D)$: **three** constants.
Solve for y. Apply $(D+1)$ to (3) and $-(D-4)$ to (4), and add:

$$\big[D^2(D+1) - D(D-4)\big]y = (D+1)t^2 \implies (D^3 + 4D)\,y = t^2 + 2t. \qquad (5)$$

$m(m^2+4) = 0$ gives $m = 0, \pm 2i$, so $y_c = c_1 + c_2\sin 2t + c_3\cos 2t$. Because $m = 0$ is a root, the polynomial trial is multiplied by t: $y_p = At + Bt^2 + Ct^3$. Then $(6C + 4A) + 8Bt + 12Ct^2 = 2t + t^2$, so $C = \frac{1}{12}$, $B = \frac{1}{4}$, $A = -\frac{1}{8}$:

$$y = c_1 + c_2\sin 2t + c_3\cos 2t + \tfrac{1}{12}t^3 + \tfrac{1}{4}t^2 - \tfrac{1}{8}t.$$

Solve for x. (3) minus $D(4)$ gives $-(D^2+4)x = t^2$. So $x_c = c_4\sin 2t + c_5\cos 2t$, and $x_p = A + Bt + Ct^2$ gives $C = -\frac{1}{4}$, $B = 0$, $A = \frac{1}{8}$:

$$x = c_4\sin 2t + c_5\cos 2t + \tfrac{1}{8} - \tfrac{1}{4}t^2.$$

Reduce five constants to three. Substitute into (4), $x' + x + y' = 0$. The polynomial terms cancel, leaving

$$(2c_4 + c_5 + 2c_2)\cos 2t + (c_4 - 2c_5 - 2c_3)\sin 2t = 0 \ \ \forall t \implies c_2 = -\tfrac{1}{2}(2c_4 + c_5), \quad c_3 = \tfrac{1}{2}(c_4 - 2c_5).$$

$$\boxed{\begin{aligned} x(t) &= c_4\sin 2t + c_5\cos 2t + \tfrac{1}{8} - \tfrac{1}{4}t^2 \\ y(t) &= c_1 - \tfrac{1}{2}(2c_4 + c_5)\sin 2t + \tfrac{1}{2}(c_4 - 2c_5)\cos 2t + \tfrac{1}{12}t^3 + \tfrac{1}{4}t^2 - \tfrac{1}{8}t \end{aligned}}$$

Three constants c_1, c_4, c_5 remain. Substituting into (3) gives an identity — a free check.

4. The matrix method for systems of ODEs

Elimination works, but it grows painful as the number of unknowns grows. The matrix method treats all the unknowns at once, and turns the problem into one of finding eigenvalues and eigenvectors.

Normal form

A first-order system of ODEs with constant coefficients is in **normal form** when each derivative stands alone on the left:

$$\begin{aligned}
\frac{dx_1}{dt} &= a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n + f_1(t)\\
\frac{dx_2}{dt} &= a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n + f_2(t)\\
&\vdots\\
\frac{dx_n}{dt} &= a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n + f_n(t)
\end{aligned}
\qquad \text{or, briefly,} \qquad
\frac{dx_i}{dt} = \sum_{j=1}^{n} a_{ij}\,x_j + f_i(t),$$

where the a_{ij} $(i, j = 1, 2, \ldots, n)$ are given constants.

Matrix form. Collect the unknowns, their derivatives and the forcing terms into vectors:

$$\mathbf{x}(t) = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \qquad \mathbf{x}'(t) = \begin{pmatrix} dx_1/dt \\ dx_2/dt \\ \vdots \\ dx_n/dt \end{pmatrix}, \qquad \mathbf{f}(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \\ \vdots \\ f_n(t) \end{pmatrix}, \qquad A = (a_{ij})_{n\times n}.$$

The whole system then reads

$$\boxed{\mathbf{x}'(t) = A\,\mathbf{x}(t) + \mathbf{f}(t)}$$

It is **non-homogeneous** in general, and **homogeneous** if $\mathbf{f}(t) = \mathbf{0}$.

Example 4 — writing a system in matrix form

$$\frac{dx}{dt} = x + y + 4z, \qquad \frac{dy}{dt} = x - 7y + z, \qquad \frac{dz}{dt} = 2x + 3y - 2z.$$

In matrix form,

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}, \qquad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & 1 & 4 \\ 1 & -7 & 1 \\ 2 & 3 & -2 \end{pmatrix}.$$

Solving the homogeneous system

Consider the first-order homogeneous system

$$\mathbf{x}'(t) = A\,\mathbf{x}(t). \qquad (1)$$

For one equation $x' = ax$ the solution is Ce^{at}. By analogy, seek a solution of (1) of the form

$$\mathbf{x}(t) = \mathbf{u}\,e^{\lambda t}, \qquad (2)$$

where $\mathbf{u}$ is a constant vector and λ a constant, both still to be found. Substituting (2) into (1):

$$\frac{d\mathbf{x}}{dt} = \lambda\,\mathbf{u}\,e^{\lambda t} = A\,\mathbf{u}\,e^{\lambda t} \implies (A - \lambda I)\,\mathbf{u}\,e^{\lambda t} = \mathbf{0},$$

where I is the $n \times n$ identity matrix. Since $e^{\lambda t} \neq 0$,

$$(A - \lambda I)\,\mathbf{u} = \mathbf{0}.$$

This is the **eigenvalue equation**: λ is an **eigenvalue** of A and $\mathbf{u}$ a corresponding **eigenvector**. It has a non-trivial solution $\mathbf{u} \neq \mathbf{0}$ only if

$$p(\lambda) = \det(A - \lambda I) = 0, \qquad \text{the } \textbf{characteristic equation}.$$

The characteristic equation is used to find λ; once λ is known, the eigenvalue equation is used to find $\mathbf{u}$.

> **Why λI and not just λ?** λ is a number and A is a matrix, and a number cannot be subtracted from a matrix. $\lambda\mathbf{u} = \lambda I\mathbf{u}$, so we subtract the matrix λI (with λ down the diagonal) from A.

Theorem 1 — the general solution

Suppose the $n \times n$ matrix A has n linearly independent eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$. Let λ_i be the eigenvalue (real or complex, not necessarily distinct) corresponding to $\mathbf{u}_i$. Then

$$\left\{\mathbf{x}_1 = \mathbf{u}_1 e^{\lambda_1 t},\ \mathbf{x}_2 = \mathbf{u}_2 e^{\lambda_2 t},\ \ldots,\ \mathbf{x}_n = \mathbf{u}_n e^{\lambda_n t}\right\}$$

is a fundamental set of solutions of (1) on $(-\infty, \infty)$, and the general solution is

$$\mathbf{x}(t) = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t) + \cdots + c_n\mathbf{x}_n(t) = c_1\mathbf{u}_1 e^{\lambda_1 t} + c_2\mathbf{u}_2 e^{\lambda_2 t} + \cdots + c_n\mathbf{u}_n e^{\lambda_n t},$$

where the c_i are arbitrary constants. Eigenvectors of *distinct* eigenvalues are always independent, so the theorem applies whenever the eigenvalues are all different.

Recipe. (i) Solve $\det(A-\lambda I) = 0$ for the eigenvalues. (ii) For each λ solve $(A-\lambda I)\mathbf{u} = \mathbf{0}$; the rows are dependent, so a free parameter remains — choose it to clear fractions. (iii) Write $\mathbf{x} = \sum c_i \mathbf{u}_i e^{\lambda_i t}$. (iv) For an IVP, solve $\mathbf{x}(0) = \sum c_i \mathbf{u}_i$ for the constants.

Example 5 — a 3×3 system with distinct real eigenvalues

Solve $\mathbf{x}'(t) = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{pmatrix} \mathbf{x}(t)$.

Step 1: the eigenvalues.

$$p(\lambda) = |A - \lambda I| = \left| \begin{pmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \right| = \begin{vmatrix} 1-\lambda & 2 & -1 \\ 1 & -\lambda & 1 \\ 4 & 4 & 3-\lambda \end{vmatrix} = 0.$$

Method 1 — the diagonal (Sarrus) rule. Copy the first two columns to the right; add the three "down" diagonal products and subtract the three "up" ones:

$$\begin{aligned} &(1-\lambda)(-\lambda)(3-\lambda) + 2\cdot 1\cdot 4 + (-1)\cdot 1\cdot 4 - \big[4(-\lambda)(-1) + 4\cdot 1\cdot(1-\lambda) + (3-\lambda)\cdot 1\cdot 2\big] \\ &\quad = -\lambda(3 - 4\lambda + \lambda^2) + 4 - \big[4\lambda + 4 - 4\lambda + 6 - 2\lambda\big] = -\lambda^3 + 4\lambda^2 - \lambda - 6 = 0, \end{aligned}$$

i.e. $\lambda^3 - 4\lambda^2 + \lambda + 6 = 0$. Trying divisors of 6, $\lambda = -1$ is a root, and $\lambda^3 - 4\lambda^2 + \lambda + 6 = (\lambda+1)(\lambda^2 - 5\lambda + 6) = (\lambda+1)(\lambda-2)(\lambda-3)$.

Method 2 — cofactor expansion along row 1, keeping factors visible as you go:

$$\begin{aligned} &(1-\lambda)\big[-\lambda(3-\lambda) - 4\big] - 2\big[(3-\lambda) - 4\big] - 1\big[4 + 4\lambda\big] \\ &\quad = (1-\lambda)(\lambda-4)(\lambda+1) + 2(\lambda+1) - 4(\lambda+1) = (\lambda+1)\big[(1-\lambda)(\lambda-4) - 2\big] \\ &\quad = (\lambda+1)(-\lambda^2 + 5\lambda - 6) = -(\lambda+1)(\lambda-2)(\lambda-3) = 0. \end{aligned}$$

Either way, the eigenvalues are $\lambda_1 = -1,\ \lambda_2 = 2,\ \lambda_3 = 3$.

Step 2: an eigenvector for each eigenvalue.

For $\lambda = -1$*:* $(A + I)\mathbf{u} = \mathbf{0}$, i.e. $\begin{pmatrix} 2 & 2 & -1 \\ 1 & 1 & 1 \\ 4 & 4 & 4 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$:

$$2u_1 + 2u_2 - u_3 = 0 \ \Rightarrow\ u_3 = 2u_1 + 2u_2, \qquad u_1 + u_2 + u_3 = 0 \ \text{(row 3 is the same)}.$$

Substituting, $3(u_1 + u_2) = 0$, so $u_2 = -u_1$ and $u_3 = 0$:

$$\mathbf{u} = \begin{pmatrix} u_1 \\ -u_1 \\ 0 \end{pmatrix} = u_1 \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \quad \Longrightarrow \quad \mathbf{x}_1(t) = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \mathrm{e}^{-t}$$

(any non-zero multiple will do; here $u_1 = -1$).

For $\lambda = 2$*:* $(A - 2I)\mathbf{u} = \mathbf{0}$, i.e. $\begin{pmatrix} -1 & 2 & -1 \\ 1 & -2 & 1 \\ 4 & 4 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \mathbf{0}$:

$$u_1 - 2u_2 + u_3 = 0 \ \Rightarrow\ u_3 = 2u_2 - u_1, \qquad 4u_1 + 4u_2 + u_3 = 0.$$

Substituting, $4u_1 + 4u_2 + 2u_2 - u_1 = 3u_1 + 6u_2 = 0$, so $u_1 = -2u_2$ and $u_3 = 4u_2$:

$$\mathbf{u} = u_2 \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} \quad \Longrightarrow \quad \mathbf{x}_2(t) = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} \mathrm{e}^{2t}.$$

For $\lambda = 3$*:* $(A - 3I)\mathbf{u} = \mathbf{0}$, i.e. $\begin{pmatrix} -2 & 2 & -1 \\ 1 & -3 & 1 \\ 4 & 4 & 0 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \mathbf{0}$:

$$-2u_1 + 2u_2 - u_3 = 0, \qquad u_1 - 3u_2 + u_3 = 0, \qquad 4u_1 + 4u_2 = 0 \ \Rightarrow\ u_1 = -u_2.$$

Substituting into the first, $2u_2 + 2u_2 - u_3 = 0$, so $u_3 = 4u_2$:

$$\mathbf{u} = u_2 \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix} \quad \Longrightarrow \quad \mathbf{x}_3(t) = \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix} \mathrm{e}^{3t}.$$

Check: $A(-2,1,4)^{\mathsf{T}} = (-4,2,8)^{\mathsf{T}} = 2(-2,1,4)^{\mathsf{T}}$ ✓, $A(-1,1,4)^{\mathsf{T}} = (-3,3,12)^{\mathsf{T}} = 3(-1,1,4)^{\mathsf{T}}$ ✓.

Step 3: the general solution.

$$\boxed{\mathbf{x}(t) = c_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \mathrm{e}^{-t} + c_2 \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} \mathrm{e}^{2t} + c_3 \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix} \mathrm{e}^{3t}}$$

Check every eigenvector. Compute $A\mathbf{u}$ and confirm it equals $\lambda\mathbf{u}$. If a calculation forces $\mathbf{u} = \mathbf{0}$, you have made an arithmetic slip: an eigenvalue *always* has a non-zero eigenvector, so go back and recheck the rows.

Example 6 — the same system as an initial-value problem

Solve $\mathbf{x}'(t) = \begin{pmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{pmatrix} \mathbf{x}(t), \quad \mathbf{x}(0) = \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$.

From Example 5, the general solution is $\mathbf{x}(t) = c_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \mathrm{e}^{-t} + c_2 \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} \mathrm{e}^{2t} + c_3 \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix} \mathrm{e}^{3t}$.

N.B. — the matrix way of writing it. The three solutions can be placed side by side as the columns of a matrix:

$$\mathbf{x}(t) = \underbrace{\begin{pmatrix} -\mathrm{e}^{-t} & -2\mathrm{e}^{2t} & -\mathrm{e}^{3t} \\ \mathrm{e}^{-t} & \mathrm{e}^{2t} & \mathrm{e}^{3t} \\ 0 & 4\mathrm{e}^{2t} & 4\mathrm{e}^{3t} \end{pmatrix}}_{X(t)} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = X(t)\,\mathbf{c}.$$

$X(t)$ is the **fundamental matrix** (see §6). Fitting the initial condition means solving $X(0)\mathbf{c} = \mathbf{x}(0)$.

Fitting $\mathbf{x}(0)$. At $t = 0$ every exponential equals 1:

$$c_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} + c_3 \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} \quad \Longrightarrow \quad \begin{cases} -c_1 - 2c_2 - c_3 = -1 & (*) \\ c_1 + c_2 + c_3 = -1 & (**) \\ 4c_2 + 4c_3 = 0 & (***) \end{cases}$$

From $(***)$: $c_3 = -c_2$. Then $(**)$ gives $c_1 = -1$, and $(*)$ gives $1 - 2c_2 + c_2 = -1$, so $c_2 = 2$ and $c_3 = -2$.

$$\boxed{\mathbf{x}(t) = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \mathrm{e}^{-t} + 2 \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} \mathrm{e}^{2t} - 2 \begin{pmatrix} -1 \\ 1 \\ 4 \end{pmatrix} \mathrm{e}^{3t}}$$

Check: $\mathbf{x}(0) = (1 - 4 + 2,\ -1 + 2 - 2,\ 0 + 8 - 8)^{\mathsf{T}} = (-1, -1, 0)^{\mathsf{T}}$ ✓.

Example 7 — a quick 2×2 system

Solve $\mathbf{x}' = \begin{pmatrix} 2 & -5 \\ 1 & -4 \end{pmatrix} \mathbf{x}$, i.e. $x_1' = 2x_1 - 5x_2$, $x_2' = x_1 - 4x_2$.

Eigenvalues. $\begin{vmatrix} 2 - \lambda & -5 \\ 1 & -4 - \lambda \end{vmatrix} = \lambda^2 + 2\lambda - 3 = (\lambda + 3)(\lambda - 1) = 0$, so $\lambda_1 = 1$, $\lambda_2 = -3$.

Eigenvectors. $\lambda = 1$: both rows of $A - I$ are $(1, -5)$, so $u_1 = 5u_2$: $\mathbf{u}_1 = (5, 1)^{\mathsf{T}}$. $\lambda = -3$: rows $(5, -5)$, $(1, -1)$, so $u_1 = u_2$: $\mathbf{u}_2 = (1, 1)^{\mathsf{T}}$.

$$\boxed{\mathbf{x}(t) = c_1 \begin{pmatrix} 5 \\ 1 \end{pmatrix} \mathrm{e}^{t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \mathrm{e}^{-3t}}$$

5. Complex eigenvalues

A real matrix has complex eigenvalues in conjugate pairs $\alpha \pm i\beta$. One complex eigenpair yields *two* real solutions, so only one member of each pair needs work.

Theorem 2 — real solutions from a complex pair

Let $\lambda = \alpha + i\beta$ ($\beta > 0$) be a complex eigenvalue of the coefficient matrix A of $\mathbf{x}'(t) = A\mathbf{x}(t)$, with corresponding eigenvector $\mathbf{u} = \mathbf{a} + i\mathbf{b}$ ($\mathbf{a}, \mathbf{b}$ real). Two linearly independent real vector solutions of the system are

$$\mathbf{x}_1(t) = \mathrm{e}^{\alpha t}\big(\mathbf{a}\cos\beta t - \mathbf{b}\sin\beta t\big), \qquad \mathbf{x}_2(t) = \mathrm{e}^{\alpha t}\big(\mathbf{a}\sin\beta t + \mathbf{b}\cos\beta t\big).$$

Proof. Since A is real, conjugating $A\mathbf{u} = \lambda\mathbf{u}$ gives $A\bar{\mathbf{u}} = \bar{\lambda}\bar{\mathbf{u}}$. So two linearly independent (complex) solutions are

$$\mathbf{z}_1(t) = \mathrm{e}^{\lambda_1 t}\mathbf{u}_1 = \mathrm{e}^{(\alpha+i\beta)t}(\mathbf{a}+i\mathbf{b}), \qquad \mathbf{z}_2(t) = \mathrm{e}^{\lambda_2 t}\mathbf{u}_2 = \mathrm{e}^{(\alpha-i\beta)t}(\mathbf{a}-i\mathbf{b}).$$

Using Euler's formula $\mathrm{e}^{\pm i\theta} = \cos\theta \pm i\sin\theta$, the general solution is

$$\begin{aligned}
\mathbf{z}(t) = c_1\mathbf{z}_1 + c_2\mathbf{z}_2 &= c_1\mathrm{e}^{\alpha t}(\cos\beta t + i\sin\beta t)(\mathbf{a}+i\mathbf{b}) + c_2\mathrm{e}^{\alpha t}(\cos\beta t - i\sin\beta t)(\mathbf{a}-i\mathbf{b}) \\
&= \mathrm{e}^{\alpha t}\Big[c_1(\mathbf{a}\cos\beta t - \mathbf{b}\sin\beta t) + ic_1(\mathbf{a}\sin\beta t + \mathbf{b}\cos\beta t)\Big] \\
&\quad + \mathrm{e}^{\alpha t}\Big[c_2(\mathbf{a}\cos\beta t - \mathbf{b}\sin\beta t) - ic_2(\mathbf{a}\sin\beta t + \mathbf{b}\cos\beta t)\Big] \\
&= (c_1+c_2)\,\mathrm{e}^{\alpha t}(\mathbf{a}\cos\beta t - \mathbf{b}\sin\beta t) + i(c_1-c_2)\,\mathrm{e}^{\alpha t}(\mathbf{a}\sin\beta t + \mathbf{b}\cos\beta t) \\
&= \tilde{c}_1\,\mathrm{e}^{\alpha t}(\mathbf{a}\cos\beta t - \mathbf{b}\sin\beta t) + \tilde{c}_2\,\mathrm{e}^{\alpha t}(\mathbf{a}\sin\beta t + \mathbf{b}\cos\beta t),
\end{aligned}$$

where $\tilde{c}_1 = c_1 + c_2$ and $\tilde{c}_2 = i(c_1 - c_2)$. Thus

$$\mathbf{x}_1(t) = \mathrm{Re}\{\mathbf{z}_1(t)\} = \mathrm{e}^{\alpha t}(\mathbf{a}\cos\beta t - \mathbf{b}\sin\beta t), \qquad \mathbf{x}_2(t) = \mathrm{Im}\{\mathbf{z}_1(t)\} = \mathrm{e}^{\alpha t}(\mathbf{a}\sin\beta t + \mathbf{b}\cos\beta t).$$

Choosing $c_2 = \bar{c}_1$ makes $\tilde{c}_1, \tilde{c}_2$ real, so $\mathbf{x}_1, \mathbf{x}_2$ are real solutions, independent because $\mathbf{z}_1, \mathbf{z}_2$ are. ■

Example 8 — purely imaginary eigenvalues: a centre

Solve $\mathbf{x}'(t) = \begin{pmatrix} 2 & 8 \\ -1 & -2 \end{pmatrix}\mathbf{x}(t)$.

Eigenvalues. The characteristic equation is

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 8 \\ -1 & -2-\lambda \end{vmatrix} = (2-\lambda)(-2-\lambda) + 8 = -4 - 2\lambda + 2\lambda + \lambda^2 + 8 = \lambda^2 + 4 = 0,$$

so $\lambda = \pm 2i$. Take the eigenvalue with *positive* imaginary part: $\lambda = 2i$, so $\alpha = 0$, $\beta = 2$.

Eigenvector for $\lambda = 2i$.

$$\begin{pmatrix} 2-2i & 8 \\ -1 & -2-2i \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \begin{cases} (2-2i)u_1 + 8u_2 = 0 \\ -u_1 + (-2-2i)u_2 = 0 \;\Rightarrow\; u_1 = (-2-2i)u_2. \end{cases}$$

(The first row gives the same relation: $(2-2i)(-2-2i) = -8$.) With $u_2 = 1$:

$$\mathbf{u} = \begin{pmatrix} -2-2i \\ 1 \end{pmatrix} = \underbrace{\begin{pmatrix} -2 \\ 1 \end{pmatrix}}_{\mathbf{a}} + i\underbrace{\begin{pmatrix} -2 \\ 0 \end{pmatrix}}_{\mathbf{b}}.$$

Real solutions ($\alpha = 0$, $\beta = 2$):

$$\mathbf{x}_1(t) = \mathrm{e}^{0t}\left\{\begin{pmatrix} -2 \\ 1 \end{pmatrix}\cos 2t - \begin{pmatrix} -2 \\ 0 \end{pmatrix}\sin 2t\right\} = \begin{pmatrix} -2\cos 2t + 2\sin 2t \\ \cos 2t \end{pmatrix},$$

$$\mathbf{x}_2(t) = \mathrm{e}^{0t}\left\{\begin{pmatrix} -2 \\ 1 \end{pmatrix}\sin 2t + \begin{pmatrix} -2 \\ 0 \end{pmatrix}\cos 2t\right\} = \begin{pmatrix} -2\sin 2t - 2\cos 2t \\ \sin 2t \end{pmatrix}.$$

$$\boxed{\mathbf{x}(t) = c_1 \begin{pmatrix} -2\cos 2t + 2\sin 2t \\ \cos 2t \end{pmatrix} + c_2 \begin{pmatrix} -2\sin 2t - 2\cos 2t \\ \sin 2t \end{pmatrix}}$$

Check of $\mathbf{x}_1$*:* $\mathbf{x}_1' = (4\sin 2t + 4\cos 2t,\ -2\sin 2t)^{\mathsf{T}}$ and $A\mathbf{x}_1 = (4\cos 2t + 4\sin 2t,\ -2\sin 2t)^{\mathsf{T}}$ ✓. With $\alpha = 0$ every solution is periodic: the origin is a **centre**.

What if we use $\lambda = -2i$? Its eigenvector is the conjugate, $\bar{\mathbf{u}} = (-2+2i,\ 1)^{\mathsf{T}}$, so $\mathbf{b}$ changes sign to $(2,0)^{\mathsf{T}}$ *and* $\beta = -2$. Putting $\mathbf{b} = (2,0)^{\mathsf{T}}$ into Theorem 2 with $\beta = +2$ gives $(-2\cos 2t - 2\sin 2t,\ \cos 2t)^{\mathsf{T}}$, which is *not* a solution. Either always work with $\alpha + i\beta$, $\beta > 0$, or expand $\mathbf{u}\,\mathrm{e}^{\lambda t}$ directly and take real and imaginary parts (Example 10); both reproduce the answer above.

Example 9 — $\alpha \neq 0$: a spiral

Solve $\mathbf{x}' = \begin{pmatrix} 4 & 5 \\ -2 & 6 \end{pmatrix}\mathbf{x}$.

Eigenvalues. $\lambda^2 - 10\lambda + 34 = 0$, so $\lambda = \dfrac{10 \pm \sqrt{100 - 136}}{2} = 5 \pm 3i$: $\alpha = 5$, $\beta = 3$.

Eigenvector for $\lambda = 5+3i$. Row 1 of $A - \lambda I$: $(-1-3i)u_1 + 5u_2 = 0$, so $u_1 = \dfrac{5u_2}{1+3i} = \frac{1}{2}(1-3i)u_2$. Taking $u_2 = 2$ clears fractions:

$$\mathbf{u} = \begin{pmatrix} 1 - 3i \\ 2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}_{\mathbf{a}} + i \underbrace{\begin{pmatrix} -3 \\ 0 \end{pmatrix}}_{\mathbf{b}}.$$

$$\boxed{\mathbf{x}(t) = c_1 \mathrm{e}^{5t} \begin{pmatrix} \cos 3t + 3\sin 3t \\ 2\cos 3t \end{pmatrix} + c_2 \mathrm{e}^{5t} \begin{pmatrix} \sin 3t - 3\cos 3t \\ 2\sin 3t \end{pmatrix}}$$

A real rescaling of $\mathbf{u}$ (e.g. $u_2 = 1$) only rescales c_1, c_2. Since $\alpha = 5 > 0$, trajectories spiral outwards — an **unstable spiral**.

Example 10 — straight from Euler's formula

Solve $\mathbf{x}' = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}\mathbf{x}$.

$(1-\lambda)^2 + 4 = 0 \Rightarrow \lambda^2 - 2\lambda + 5 = 0 \Rightarrow \lambda = 1 \pm 2i$. For $\lambda = 1 + 2i$: $A - \lambda I = \begin{pmatrix} -2i & 2 \\ -2 & -2i \end{pmatrix}$, row 1 gives $u_2 = iu_1$, so $\mathbf{u} = (1, i)^{\mathsf{T}}$. Expanding,

$$\mathbf{u}\,\mathrm{e}^{(1+2i)t} = \mathrm{e}^t \begin{pmatrix} 1 \\ i \end{pmatrix} (\cos 2t + i\sin 2t) = \mathrm{e}^t \begin{pmatrix} \cos 2t \\ -\sin 2t \end{pmatrix} + i\,\mathrm{e}^t \begin{pmatrix} \sin 2t \\ \cos 2t \end{pmatrix}.$$

The real and imaginary parts are the two real solutions:

$$\boxed{\mathbf{x}(t) = c_1 \mathrm{e}^t \begin{pmatrix} \cos 2t \\ -\sin 2t \end{pmatrix} + c_2 \mathrm{e}^t \begin{pmatrix} \sin 2t \\ \cos 2t \end{pmatrix}}$$

6. Some concepts from matrix theory

When an eigenvalue is repeated, Theorem 1 may not supply enough solutions. To find the missing ones we need the matrix exponential.

Definition — the matrix exponential

If A is a square matrix, the **matrix exponential** is

$$\mathrm{e}^{A} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots + \frac{A^n}{n!} + \cdots = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

Useful property: $\mathrm{e}^{A+B} = \mathrm{e}^{A}\mathrm{e}^{B}$ *provided* $AB = BA$.

The proviso matters: for numbers $ab = ba$ always, but for matrices it usually fails. Two matrices that *always* commute with A are I and λI — this is what makes §7 work.

Proposition — e^{At} solves the system

$\dfrac{d}{dt}\left(\mathrm{e}^{At}\right) = A\,\mathrm{e}^{At}$. Hence e^{At} is a (matrix) solution of $\mathbf{x}'(t) = A\mathbf{x}(t)$ (1).

Proof. By definition, $\mathrm{e}^{At} = I + At + \dfrac{A^2t^2}{2!} + \cdots + \dfrac{A^nt^n}{n!} + \cdots$. Differentiating term by term,

$$\frac{d}{dt}\left[\mathrm{e}^{At}\right] = A + \frac{2A^2t}{2!} + \frac{3A^3t^2}{3!} + \cdots + \frac{nA^nt^{n-1}}{n!} + \cdots = A + \frac{A^2t}{1!} + \frac{A^3t^2}{2!} + \cdots + \frac{A^nt^{n-1}}{(n-1)!} + \cdots$$

$$= A\left[I + \frac{At}{1!} + \frac{A^2t^2}{2!} + \cdots + \frac{A^{n-1}t^{n-1}}{(n-1)!} + \cdots\right] = A\,\mathrm{e}^{At}. \qquad \blacksquare$$

Fundamental matrix and generalised eigenvectors

- e^{At} is a **fundamental matrix** of (1): its n columns are linearly independent solutions of (1). Consequently $\mathrm{e}^{At}\mathbf{u}$ is a solution for *every* constant vector $\mathbf{u}$.
- More generally, any matrix $X(t)$ whose columns are n linearly independent solutions of (1) is a fundamental matrix, and the general solution is $\mathbf{x} = X(t)\,\mathbf{c}$ (as in Example 6).
- If λ is an eigenvalue of $A_{n\times n}$, a non-zero vector $\mathbf{u}$ with

  $$(A - \lambda I)^k\,\mathbf{u} = \mathbf{0}$$

  for some natural number k is called a **generalised eigenvector** associated with λ. (For $k = 1$ this is an ordinary eigenvector.)
- **Fact** (from the Cayley–Hamilton theorem): if λ has multiplicity m, then $(A - \lambda I)^m\mathbf{u} = \mathbf{0}$ has m linearly independent solutions $\mathbf{u}$. So a repeated eigenvalue always supplies as many generalised eigenvectors as its multiplicity.

Example 11 — the exponential of a simple matrix

Find e^{At} for $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$.

Write $A = 2I + N$ with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Since $2I$ commutes with N, $\mathrm{e}^{At} = \mathrm{e}^{2It}\mathrm{e}^{Nt}$. Here $\mathrm{e}^{2It} = \mathrm{e}^{2t}I$

and, because $N^2 = 0$, the series for e^{Nt} stops after two terms: $\mathrm{e}^{Nt} = I + Nt$. Therefore

$$\boxed{\mathrm{e}^{At} = \mathrm{e}^{2t}\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}}$$

Its columns $(1,0)^{\mathsf{T}}\mathrm{e}^{2t}$ and $(t,1)^{\mathsf{T}}\mathrm{e}^{2t}$ solve $\mathbf{x}' = A\mathbf{x}$, even though A has only one eigenvector. That t in the second column is exactly what the repeated-eigenvalue method produces.

7. Repeated eigenvalues: generalised eigenvectors

The key expansion. For any eigenvalue λ, since λI commutes with A,

$$\begin{aligned}\mathrm{e}^{At}\mathbf{u} &= \mathrm{e}^{(A-\lambda I+\lambda I)t}\mathbf{u} = \mathrm{e}^{\lambda I t}\,\mathrm{e}^{(A-\lambda I)t}\mathbf{u} = \mathrm{e}^{\lambda t}\,\mathrm{e}^{(A-\lambda I)t}\mathbf{u} \\ &= \mathrm{e}^{\lambda t}\Big[I + (A-\lambda I)t + \frac{(A-\lambda I)^2 t^2}{2!} + \frac{(A-\lambda I)^3 t^3}{3!} + \cdots\Big]\mathbf{u}.\end{aligned}$$

This is a solution of (1) for every $\mathbf{u}$ (Proposition). If $\mathbf{u}$ is a generalised eigenvector with $(A-\lambda I)^k\mathbf{u} = \mathbf{0}$, every term from $(A-\lambda I)^k$ onwards vanishes and the infinite series becomes a *finite* sum.

Solutions associated with an eigenvalue λ of multiplicity m

$$\begin{aligned}
\mathbf{x}_1(t) &= \mathbf{u}_1\,\mathrm{e}^{\lambda t}, && \text{where } (A-\lambda I)\mathbf{u}_1 = \mathbf{0},\\
\mathbf{x}_2(t) &= \big[I + (A-\lambda I)t\big]\mathbf{u}_2\,\mathrm{e}^{\lambda t}, && \text{where } (A-\lambda I)^2\mathbf{u}_2 = \mathbf{0},\\
\mathbf{x}_3(t) &= \Big[I + (A-\lambda I)t + \frac{(A-\lambda I)^2 t^2}{2!}\Big]\mathbf{u}_3\,\mathrm{e}^{\lambda t}, && \text{where } (A-\lambda I)^3\mathbf{u}_3 = \mathbf{0},\\
&\ \ \vdots \\
\mathbf{x}_k(t) &= \Big[I + (A-\lambda I)t + \cdots + \frac{(A-\lambda I)^{k-1}t^{k-1}}{(k-1)!}\Big]\mathbf{u}_k\,\mathrm{e}^{\lambda t}, && \text{where } (A-\lambda I)^k\mathbf{u}_k = \mathbf{0},
\end{aligned}$$

for $k = 1, \ldots, m$, choosing each new $\mathbf{u}_k$ linearly independent of the ones already used.

Order of work. Deal with the non-repeated eigenvalues first (one eigenvector each). For a repeated eigenvalue, find its ordinary eigenvectors; only if there are fewer than its multiplicity do you go on to $(A-\lambda I)^2\mathbf{u} = \mathbf{0}$, and so on.

Example 12 — a double eigenvalue with only one eigenvector

Use generalised eigenvectors (where applicable) to find the general solution of $\mathbf{x}'(t) = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix}\mathbf{x}(t)$.

Eigenvalues. Expanding along the first column,

$$\begin{aligned}\det(A-\lambda I) &= \begin{vmatrix} 1-\lambda & 1 & 1 \\ 2 & 1-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{vmatrix} = (1-\lambda)\big[(1-\lambda)^2 - 1\big] - 2\big[(1-\lambda)+1\big] \\ &= -\lambda^3 + 3\lambda^2 - 4 = 0 \quad\Longleftrightarrow\quad \lambda^3 - 3\lambda^2 + 4 = 0.\end{aligned}$$

$\lambda = -1$ is a root; long division gives $\lambda^3 - 3\lambda^2 + 4 = (\lambda+1)(\lambda^2 - 4\lambda + 4) = (\lambda+1)(\lambda-2)^2$. So $\lambda = -1$ and $\lambda = 2$ (multiplicity 2).

For $\lambda = -1$ (non-repeated): $\begin{pmatrix} 2 & 1 & 1 \\ 2 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \mathbf{0}$ gives $2u_1+u_2+u_3 = 0$, $2u_1+2u_2-u_3 = 0$, $-u_2+2u_3 = 0$. So $u_2 = 2u_3$, and the first equation gives $2u_1 = -3u_3$. Taking $u_3 = 2$:

$$\mathbf{u} = \begin{pmatrix} -3 \\ 4 \\ 2 \end{pmatrix}, \qquad \mathbf{x}_1(t) = \begin{pmatrix} -3 \\ 4 \\ 2 \end{pmatrix} \mathrm{e}^{-t}. \qquad \big(\text{Check: } A\mathbf{u} = (3,-4,-2)^{\mathsf{T}} = -\mathbf{u}\ \checkmark\big)$$

For $\lambda = 2$: the first independent solution. $\begin{pmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \mathbf{0}$: $-u_1+u_2+u_3 = 0$, $2u_1-u_2-u_3 = 0$; adding, $u_1 = 0$, and then $u_2 = -u_3$:

$$\mathbf{u}_1 = \begin{pmatrix} 0 \\ -u_3 \\ u_3 \end{pmatrix} = u_3 \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \qquad \mathbf{x}_2(t) = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \mathrm{e}^{2t}.$$

Only one free parameter, so only *one* eigenvector for a double eigenvalue: we need a generalised eigenvector.

For $\lambda = 2$: the second independent solution. Solve $(A-2I)^2\mathbf{u}_2 = \mathbf{0}$:

$$(A-2I)^2 = \begin{pmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix} \begin{pmatrix} -1 & 1 & 1 \\ 2 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix} = \begin{pmatrix} 3 & -3 & -3 \\ -4 & 4 & 4 \\ -2 & 2 & 2 \end{pmatrix}.$$

Every row is a multiple of $(1,-1,-1)$, so $u_1 = u_2 + u_3$ and

$$\mathbf{u}_2 = \begin{pmatrix} u_2+u_3 \\ u_2 \\ u_3 \end{pmatrix} = u_2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + u_3 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

Choose a vector that is linearly independent of $\mathbf{u}_1 = (0,-1,1)^{\mathsf{T}}$; here either one will do. Choosing $\mathbf{u}_2 = (1,0,1)^{\mathsf{T}}$,

$$\begin{aligned} \mathbf{x}_3(t) = \big[I + (A-2I)t\big]\mathbf{u}_2\,\mathrm{e}^{2t} &= \left[\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -t & t & t \\ 2t & -t & -t \\ 0 & -t & -t \end{pmatrix}\right] \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \mathrm{e}^{2t} \\ &= \begin{pmatrix} 1-t & t & t \\ 2t & 1-t & -t \\ 0 & -t & 1-t \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \mathrm{e}^{2t} = \begin{pmatrix} 1-t+t \\ 2t-t \\ 1-t \end{pmatrix} \mathrm{e}^{2t} = \begin{pmatrix} 1 \\ t \\ 1-t \end{pmatrix} \mathrm{e}^{2t}. \end{aligned}$$

General solution.

$$\boxed{\mathbf{x}(t) = c_1 \begin{pmatrix} -3 \\ 4 \\ 2 \end{pmatrix} \mathrm{e}^{-t} + c_2 \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \mathrm{e}^{2t} + c_3 \begin{pmatrix} 1 \\ t \\ 1-t \end{pmatrix} \mathrm{e}^{2t}}$$

Check of $\mathbf{x}_3$: $\mathbf{x}_3' = (2,\ 1+2t,\ 1-2t)^{\mathsf{T}}\mathrm{e}^{2t}$ and $A(1,t,1-t)^{\mathsf{T}} = (1+t+1-t,\ 2+t-1+t,\ -t+1-t)^{\mathsf{T}} = (2,\ 1+2t,\ 1-2t)^{\mathsf{T}}$ ✓.

A shortcut: chains. Instead of computing $(A-\lambda I)^2$, solve $(A-\lambda I)\mathbf{v}=\mathbf{u}_1$ using the eigenvector already found. Then $\mathbf{x}=(\mathbf{v}+t\,\mathbf{u}_1)\mathrm{e}^{\lambda t}$ is the second solution. In the example, $\mathbf{v}=(-1,-1,0)^{\mathsf{T}}$ gives $(-1,\,-1-t,\,t)^{\mathsf{T}}\mathrm{e}^{2t}=\mathbf{x}_2-\mathbf{x}_3$ — the same solution space. For a triple eigenvalue, continue the chain: $(A-\lambda I)\mathbf{w}=\mathbf{v}$ gives $\big(\mathbf{w}+t\mathbf{v}+\frac{t^2}{2}\mathbf{u}_1\big)\mathrm{e}^{\lambda t}$.

Contrast with Unit 2. A repeated root m of an auxiliary equation gave e^{mt} and $t\mathrm{e}^{mt}$. For systems the second solution is *not* just $t\,\mathbf{u}_1\mathrm{e}^{\lambda t}$; it is $(\mathbf{v}+t\mathbf{u}_1)\mathrm{e}^{\lambda t}$, and the extra vector $\mathbf{v}$ is essential.

8. From one higher-order equation to a first-order system

Any nth-order ODE solved for its highest derivative,

$$y^{(n)}=f\big(t,y,y',\dots,y^{(n-1)}\big),\qquad(1)$$

becomes n first-order equations by naming each lower derivative:

$$x_1=y,\quad x_2=y',\quad\dots,\quad x_n=y^{(n-1)},$$

so that

$$x_1'=x_2,\quad x_2'=x_3,\quad\dots,\quad x_{n-1}'=x_n,\quad x_n'=f(t,x_1,\dots,x_n).$$

Every method of §4–7 therefore applies to single higher-order equations too. This matters because matrix methods, numerical solvers and the existence–uniqueness theorem are all stated for first-order systems.

The companion matrix

For the linear equation $y^{(n)}+a_{n-1}y^{(n-1)}+\cdots+a_1y'+a_0y=g(t)$, the system is $\mathbf{x}'=A\mathbf{x}+\mathbf{f}(t)$ with

$$A=\begin{pmatrix}0&1&0&\cdots&0\\0&0&1&\cdots&0\\\vdots&&&\ddots&\vdots\\0&0&0&\cdots&1\\-a_0&-a_1&-a_2&\cdots&-a_{n-1}\end{pmatrix},\qquad\mathbf{f}(t)=\begin{pmatrix}0\\\vdots\\0\\g(t)\end{pmatrix}.$$

Ones on the superdiagonal, negated coefficients in the last row, forcing in the last entry only. Initial data carry over: $\mathbf{x}(t_0)=\big(y(t_0),y'(t_0),\dots,y^{(n-1)}(t_0)\big)^{\mathsf{T}}$.

Example 13 — a third-order equation

Convert $y'''-y'+y=\cos t$ into a first-order system in normal form.
Solve for the top derivative: $y'''=y'-y+\cos t$. With $x_1=y$, $x_2=y'$, $x_3=y''$: $x_1'=x_2$, $x_2'=x_3$, $x_3'=x_2-x_1+\cos t$. Hence

$$\mathbf{x}'(t)=\begin{pmatrix}0&1&0\\0&0&1\\-1&1&0\end{pmatrix}\mathbf{x}(t)+\begin{pmatrix}0\\0\\\cos t\end{pmatrix}.$$

Example 14 — convert, then solve

Convert $y'' + 3y' + 2y = 0$, $y(0) = 1$, $y'(0) = 3$ into a first-order system and hence solve it.
Conversion. With $x_1 = y$, $x_2 = y'$: $x_1' = x_2$, $x_2' = -2x_1 - 3x_2$, so

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} \mathbf{x}, \qquad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Eigenvalues. $\det(A - \lambda I) = \lambda(\lambda + 3) + 2 = (\lambda + 1)(\lambda + 2) = 0$, so $\lambda = -1, -2$ — exactly the roots of the auxiliary equation of the original ODE, as always for a companion matrix.
Eigenvectors (method in §4). $\lambda = -1$: $(A + I)\mathbf{u} = \mathbf{0}$ gives $u_1 + u_2 = 0$, so $\mathbf{u}_1 = (1, -1)^{\mathsf{T}}$. $\lambda = -2$: $2u_1 + u_2 = 0$, so $\mathbf{u}_2 = (1, -2)^{\mathsf{T}}$.

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} \mathrm{e}^{-t} + c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} \mathrm{e}^{-2t}.$$

Initial data. $c_1 + c_2 = 1$, $-c_1 - 2c_2 = 3$; adding, $c_2 = -4$ and $c_1 = 5$:

$$\boxed{\mathbf{x}(t) = \begin{pmatrix} 5\mathrm{e}^{-t} - 4\mathrm{e}^{-2t} \\ -5\mathrm{e}^{-t} + 8\mathrm{e}^{-2t} \end{pmatrix}}$$

The first component is $y(t) = 5\mathrm{e}^{-t} - 4\mathrm{e}^{-2t}$; the second is $y'(t)$. *Check:* $y(0) = 1$, $y'(0) = -5 + 8 = 3$ ✓.

9. Non-homogeneous systems I: undetermined coefficients

The general solution of $\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t)$ is

$$\mathbf{x}(t) = \mathbf{x}_h(t) + \mathbf{x}_p(t),$$

where $\mathbf{x}_h$ solves the homogeneous system (§4–7) and $\mathbf{x}_p$ is any one particular solution.

When $\mathbf{f}(t)$ has finitely many independent derivatives — vectors built from polynomials, exponentials, sines and cosines, hyperbolic functions, and their sums and products — guess $\mathbf{x}_p$ of the same form with *unknown constant vectors*, substitute, and match coefficients.

Trial forms for $\mathbf{x}_p$

Forcing $\mathbf{f}(t)$	**Trial $\mathbf{x}_p(t)$**
$\mathbf{P}_n(t) = \mathbf{q}_0 + \mathbf{q}_1 t + \cdots + \mathbf{q}_n t^n$	$\mathbf{a}_0 + \mathbf{a}_1 t + \cdots + \mathbf{a}_{n+s} t^{n+s}$
$\mathbf{q}\,\mathrm{e}^{\alpha t}$	$(\mathbf{a}_0 + \mathbf{a}_1 t + \cdots + \mathbf{a}_s t^s)\,\mathrm{e}^{\alpha t}$
$\mathbf{q}\cos\beta t + \mathbf{r}\sin\beta t$	$\mathbf{a}\cos\beta t + \mathbf{b}\sin\beta t$ (times t^s-polynomials)
$\mathbf{q}\cosh\beta t + \mathbf{r}\sinh\beta t$	$\mathbf{a}\cosh\beta t + \mathbf{b}\sinh\beta t$ (likewise)
$\mathbf{P}_n(t)\cos\beta t + \mathbf{Q}_m(t)\sin\beta t$	$\mathbf{A}_N(t)\cos\beta t + \mathbf{B}_N(t)\sin\beta t$, $N = \max\{m, n\} + s$

Here $s \geq 0$ is the smallest integer such that no term of $\mathbf{x}_p$ duplicates $\mathbf{x}_h$; in practice s is the multiplicity of α (or $i\beta$, or 0 for polynomials) as an eigenvalue of A.

Systems differ from single equations here. For one ODE, resonance meant multiplying the trial by t^s. For systems you must keep *all* the lower powers too: if α is a simple eigenvalue, try $(\mathbf{a}\,t + \mathbf{b})\mathrm{e}^{\alpha t}$, not just $\mathbf{a}\,t\,\mathrm{e}^{\alpha t}$ (Example 16).

Example 15 — polynomial plus exponential forcing

Solve $\mathbf{x}' = \begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix}\mathbf{x} + \begin{pmatrix} \mathrm{e}^{2t} \\ 1 \end{pmatrix}$, $\mathbf{x}(0) = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$.

Homogeneous part. $\det(A - \lambda I) = -(2-\lambda)(2+\lambda) + 3 = \lambda^2 - 1 = 0$, so $\lambda = \pm 1$. $\lambda = -1$: $A + I$ has rows $(3, -3), (1, -1)$, so $\mathbf{u} = (1, 1)^{\mathsf{T}}$. $\lambda = 1$: rows $(1, -3)$ twice, so $\mathbf{u} = (3, 1)^{\mathsf{T}}$.

$$\mathbf{x}_h = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \mathrm{e}^{-t} + c_2 \begin{pmatrix} 3 \\ 1 \end{pmatrix} \mathrm{e}^{t}.$$

Trial. Split the forcing: $\mathbf{f} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \mathrm{e}^{2t}$. Neither 0 nor 2 is an eigenvalue, so $s = 0$: try $\mathbf{x}_p = \mathbf{a} + \mathbf{b}\,\mathrm{e}^{2t}$, with $\mathbf{x}_p' = 2\mathbf{b}\,\mathrm{e}^{2t}$. Substituting,

$$2\mathbf{b}\,\mathrm{e}^{2t} = A\mathbf{a} + A\mathbf{b}\,\mathrm{e}^{2t} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \mathrm{e}^{2t}.$$

Coefficients of e^{2t}: $(A - 2I)\mathbf{b} = -\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, with $A - 2I = \begin{pmatrix} 0 & -3 \\ 1 & -4 \end{pmatrix}$, $\det = 3$:

$$\mathbf{b} = \tfrac{1}{3}\begin{pmatrix} -4 & 3 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4/3 \\ 1/3 \end{pmatrix}.$$

Constant terms: $A\mathbf{a} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$. Here $\det A = -1$ and $A^{-1} = A$ (check: $A^2 = I$), so $\mathbf{a} = A\begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$.

$$\mathbf{x}_p = \begin{pmatrix} 3 \\ 2 \end{pmatrix} + \begin{pmatrix} 4/3 \\ 1/3 \end{pmatrix} \mathrm{e}^{2t}.$$

Initial data. $\mathbf{x}(0) = c_1(1,1)^{\mathsf{T}} + c_2(3,1)^{\mathsf{T}} + (3,2)^{\mathsf{T}} + (\tfrac{4}{3}, \tfrac{1}{3})^{\mathsf{T}} = (-1, 0)^{\mathsf{T}}$:

$$c_1 + 3c_2 = -\tfrac{16}{3}, \qquad c_1 + c_2 = -\tfrac{7}{3} \implies c_2 = -\tfrac{3}{2}, \quad c_1 = -\tfrac{5}{6}.$$

$$\boxed{\mathbf{x}(t) = -\tfrac{5}{6}\begin{pmatrix} 1 \\ 1 \end{pmatrix}\mathrm{e}^{-t} - \tfrac{3}{2}\begin{pmatrix} 3 \\ 1 \end{pmatrix}\mathrm{e}^{t} + \begin{pmatrix} 3 \\ 2 \end{pmatrix} + \begin{pmatrix} 4/3 \\ 1/3 \end{pmatrix}\mathrm{e}^{2t}}$$

Check: first component at $t = 0$: $-\tfrac{5}{6} - \tfrac{9}{2} + 3 + \tfrac{4}{3} = -1$ ✓.

Example 16 — resonance: forcing at an eigenvalue

Find a particular solution of $\mathbf{x}' = \begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix}\mathbf{x} + \begin{pmatrix} \mathrm{e}^{t} \\ 0 \end{pmatrix}$.

Now $\alpha = 1$ *is* an eigenvalue (simple), so $s = 1$ and the trial is $\mathbf{x}_p = (\mathbf{a}\,t + \mathbf{b})\mathrm{e}^{t}$, with $\mathbf{x}_p' = (\mathbf{a}\,t + \mathbf{a} + \mathbf{b})\mathrm{e}^{t}$. Matching

$$te^{t}:\ (A - I)\mathbf{a} = \mathbf{0}, \qquad \mathrm{e}^{t}:\ (A - I)\mathbf{b} = \mathbf{a} - \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The first says $\mathbf{a} = k(3, 1)^{\mathsf{T}}$. Both rows of $A - I$ are $(1, -3)$, so the second is solvable only if both entries of $\mathbf{a} - (1, 0)^{\mathsf{T}} = (3k - 1,\ k)^{\mathsf{T}}$ are equal: $3k - 1 = k$, $k = \tfrac{1}{2}$. Then $b_1 - 3b_2 = \tfrac{1}{2}$; take

$\mathbf{b} = (\frac{1}{2}, 0)^{\mathsf{T}}$.

$$\boxed{\mathbf{x}_p(t) = \begin{pmatrix} \frac{3}{2}t + \frac{1}{2} \\ \frac{1}{2}t \end{pmatrix} \mathrm{e}^t}$$

The trial $\mathbf{a}\, t\, \mathrm{e}^t$ alone would have failed. Any other choice of $\mathbf{b}$ differs by a multiple of $(3,1)^{\mathsf{T}}\mathrm{e}^t$, which is absorbed into $\mathbf{x}_h$.

10. Non-homogeneous systems II: variation of parameters

Variation of parameters needs no guessing and works for *any* continuous forcing — including $\tan t$, $\sec t$ or $\ln t$, where undetermined coefficients fails.

Fundamental matrix

If $\mathbf{x}_1, \dots, \mathbf{x}_n$ are linearly independent solutions of $\mathbf{x}' = A\mathbf{x}$, the **fundamental matrix** (§6) is $X(t) = \big[\, \mathbf{x}_1 \,\vdots\, \mathbf{x}_2 \,\vdots\, \cdots \,\vdots\, \mathbf{x}_n \,\big]$. Then:

- $X' = \big[A\mathbf{x}_1 \,\vdots\, \cdots \,\vdots\, A\mathbf{x}_n\big] = AX$: X itself solves the system;
- $\mathbf{x}_h = c_1\mathbf{x}_1 + \cdots + c_n\mathbf{x}_n = X(t)\,\mathbf{c}$, with $\mathbf{c} = (c_1, \dots, c_n)^{\mathsf{T}}$;
- $\det X(t) \neq 0$ for all t, so $X^{-1}(t)$ exists.

Derivation. For $\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t)$, replace the constant vector $\mathbf{c}$ by a function:

$$\mathbf{x}_p(t) = X(t)\,\mathbf{u}(t).$$

Then $\mathbf{x}_p' = X'\mathbf{u} + X\mathbf{u}' = AX\mathbf{u} + X\mathbf{u}'$. Substituting into the system, $AX\mathbf{u} + X\mathbf{u}' = AX\mathbf{u} + \mathbf{f}$, so

$$X(t)\,\mathbf{u}'(t) = \mathbf{f}(t) \implies \mathbf{u}' = X^{-1}\mathbf{f} \implies \mathbf{u}(t) = \int X^{-1}(t)\,\mathbf{f}(t)\,dt.$$

Variation of parameters — the formulas

$$\mathbf{x}_p(t) = X(t)\int X^{-1}(t)\,\mathbf{f}(t)\,dt, \qquad \mathbf{x}(t) = X(t)\,\mathbf{c} + X(t)\int X^{-1}(t)\,\mathbf{f}(t)\,dt.$$

For the IVP $\mathbf{x}(t_0) = \mathbf{x}_0$:

$$\mathbf{x}(t) = X(t)X^{-1}(t_0)\,\mathbf{x}_0 + X(t)\int_{t_0}^{t} X^{-1}(s)\,\mathbf{f}(s)\,ds.$$

The 2×2 inverse. For $X = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$, $X^{-1} = \dfrac{1}{ps - qr}\begin{pmatrix} s & -q \\ -r & p \end{pmatrix}$: swap the diagonal, negate the off-diagonal, divide by the determinant.

Example 17 — Example 15 again, by variation of parameters

Solve $\mathbf{x}' = \begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix}\mathbf{x} + \begin{pmatrix} \mathrm{e}^{2t} \\ 1 \end{pmatrix}$, $\mathbf{x}(0) = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$.

Fundamental matrix (from Example 15):

$$X(t) = \begin{pmatrix} \mathrm{e}^{-t} & 3\mathrm{e}^{t} \\ \mathrm{e}^{-t} & \mathrm{e}^{t} \end{pmatrix}, \qquad \det X = 1 - 3 = -2,$$

$$X^{-1}(t) = -\frac{1}{2}\begin{pmatrix} \mathrm{e}^{t} & -3\mathrm{e}^{t} \\ -\mathrm{e}^{-t} & \mathrm{e}^{-t} \end{pmatrix} = \begin{pmatrix} -\frac{1}{2}\mathrm{e}^{t} & \frac{3}{2}\mathrm{e}^{t} \\ \frac{1}{2}\mathrm{e}^{-t} & -\frac{1}{2}\mathrm{e}^{-t} \end{pmatrix}.$$

Integrand and integral.

$$X^{-1}\mathbf{f} = \begin{pmatrix} -\frac{1}{2}\mathrm{e}^{3t} + \frac{3}{2}\mathrm{e}^{t} \\ \frac{1}{2}\mathrm{e}^{t} - \frac{1}{2}\mathrm{e}^{-t} \end{pmatrix}, \qquad \int X^{-1}\mathbf{f}\,dt = \begin{pmatrix} -\frac{1}{6}\mathrm{e}^{3t} + \frac{3}{2}\mathrm{e}^{t} \\ \frac{1}{2}\mathrm{e}^{t} + \frac{1}{2}\mathrm{e}^{-t} \end{pmatrix}.$$

Particular solution.

$$\mathbf{x}_p = X\int X^{-1}\mathbf{f}\,dt = \begin{pmatrix} -\frac{1}{6}\mathrm{e}^{2t} + \frac{3}{2} + \frac{3}{2}\mathrm{e}^{2t} + \frac{3}{2} \\ -\frac{1}{6}\mathrm{e}^{2t} + \frac{3}{2} + \frac{1}{2}\mathrm{e}^{2t} + \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 4/3 \\ 1/3 \end{pmatrix}\mathrm{e}^{2t} + \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

This is exactly the $\mathbf{x}_p$ of Example 15, now found without guessing. The initial data give the same $c_1 = -\frac{5}{6}$, $c_2 = -\frac{3}{2}$, hence the same boxed solution.

Which method? Undetermined coefficients is faster when $\mathbf{f}$ is a polynomial, exponential or sinusoid and the eigenvalues are simple. Variation of parameters always works, needs no resonance bookkeeping, and is the only choice for forcings like $\sec t$ or $1/t$ — at the cost of a matrix inverse and an integral.

11. Reference: eigenvalue cases at a glance

Solutions of $\mathbf{x}' = A\mathbf{x}$

Eigenvalues	Solutions contributed	Behaviour near 0
real, distinct λ_i	$\mathbf{u}_i\,\mathrm{e}^{\lambda_i t}$	node (same signs), saddle (mixed)
complex $\alpha \pm i\beta$	$\mathrm{e}^{\alpha t}(\mathbf{a}\cos\beta t - \mathbf{b}\sin\beta t)$, $\mathrm{e}^{\alpha t}(\mathbf{a}\sin\beta t + \mathbf{b}\cos\beta t)$	centre ($\alpha = 0$), spiral ($\alpha \neq 0$)
repeated, enough eigenvectors	$\mathbf{u}_1\mathrm{e}^{\lambda t}$, $\mathbf{u}_2\mathrm{e}^{\lambda t}, \ldots$	star node
repeated, defective	$\mathbf{u}_1\mathrm{e}^{\lambda t}$, $(\mathbf{v} + t\mathbf{u}_1)\mathrm{e}^{\lambda t}$, …	improper node

Stability: all solutions decay to $\mathbf{0}$ exactly when every eigenvalue has negative real part.

12. Practice

Answers are given so you can check yourself; the method matters more than the final line.

Exercises

1. Solve by elimination: $x' = x + 2y,\ \ y' = 3x + 2y$.

2. Solve $\mathbf{x}' = \begin{pmatrix} 1 & 4 \\ 1 & 1 \end{pmatrix}\mathbf{x},\ \ \mathbf{x}(0) = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

3. Find the general real solution of $\mathbf{x}' = \begin{pmatrix} 1 & -1 \\ 5 & -3 \end{pmatrix}\mathbf{x}$.

4. Find the general solution of $\mathbf{x}' = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}\mathbf{x}$.

5. Write $y''' + 2y'' - y' - 2y = \mathrm{e}^{t}$, $y(0) = 1$, $y'(0) = 0$, $y''(0) = -1$ as a first-order IVP in normal form.

6. Use undetermined coefficients: $\mathbf{x}' = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}\mathbf{x} + \begin{pmatrix} \mathrm{e}^{2t} \\ 0 \end{pmatrix},\ \ \mathbf{x}(0) = \mathbf{0}$.

7. Use variation of parameters: $\mathbf{x}' = \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix}\mathbf{x} + \begin{pmatrix} 0 \\ \mathrm{e}^{t} \end{pmatrix}$.

Answers.

1. $x = c_1\mathrm{e}^{4t} + c_2\mathrm{e}^{-t},\ \ y = \tfrac{3}{2}c_1\mathrm{e}^{4t} - c_2\mathrm{e}^{-t}$.

2. $\mathbf{x} = \begin{pmatrix} 4\mathrm{e}^{3t} - 2\mathrm{e}^{-t} \\ 2\mathrm{e}^{3t} + \mathrm{e}^{-t} \end{pmatrix}$.

3. $\lambda = -1 \pm i$; $\mathbf{x} = c_1\mathrm{e}^{-t}\begin{pmatrix} 2\cos t - \sin t \\ 5\cos t \end{pmatrix} + c_2\mathrm{e}^{-t}\begin{pmatrix} 2\sin t + \cos t \\ 5\sin t \end{pmatrix}$.

4. $\lambda = 2$ (double, one eigenvector); $\mathbf{x} = c_1\begin{pmatrix} 1 \\ 1 \end{pmatrix}\mathrm{e}^{2t} + c_2\left[\begin{pmatrix} 1 \\ 1 \end{pmatrix}t + \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right]\mathrm{e}^{2t}$.

5. $\mathbf{x}' = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & -2 \end{pmatrix}\mathbf{x} + \begin{pmatrix} 0 \\ 0 \\ \mathrm{e}^{t} \end{pmatrix},\quad \mathbf{x}(0) = (1, 0, -1)^{\mathsf{T}}$.

6. $\mathbf{x} = \begin{pmatrix} \frac{1}{2}\mathrm{e}^{3t} - \frac{1}{3}\mathrm{e}^{2t} - \frac{1}{6}\mathrm{e}^{-t} \\ \mathrm{e}^{3t} - \frac{4}{3}\mathrm{e}^{2t} + \frac{1}{3}\mathrm{e}^{-t} \end{pmatrix}$.

7. $\mathbf{x} = c_1\begin{pmatrix} 1 \\ 2 \end{pmatrix}\mathrm{e}^{t} + c_2\begin{pmatrix} 1 \\ 1 \end{pmatrix}\mathrm{e}^{2t} + \begin{pmatrix} t+1 \\ 2t+1 \end{pmatrix}\mathrm{e}^{t}$ (any $\mathbf{x}_p$ differing by a multiple of $(1,2)^{\mathsf{T}}\mathrm{e}^{t}$ is equally correct).

End of Unit 4 — M3503, Department of Mathematics & Computer Science, NUL.